Safety Boundaries

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Teach each boundary and its specific fallback.

Supply complete toy contracts, distinguish blocked access from invalid input and preserve ordinary supported use.

Keep one habit: Ask which check failed first and what evidence or authority would permit the next step.

MAIN LESSON · SECTION 1

Prepare the main lesson and its retained extension

> Use fictional records and schedule separate extension time.

Goal and preparation

Primary target: Explain the **mechanism** and terminology of **untrusted input** boundaries

Two 45-minute sessions. **Untrusted input** must be treated according to a contract before reaching code, queries or tools. Data that resembles instructions does not acquire execution authority.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

By the end, I can explain safety boundaries using a traced example and a stated limitation.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Main plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval and readiness

Without opening the old lesson, explain one key idea from Bias and Representation. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of representation and bias evaluation

Reference example, not the only acceptable wording: Teacher explanation: Bias evaluation tests defined outcomes across relevant conditions.

Group averages can expose differences but cannot alone establish their cause or the correct remedy. Define the task, relevant groups, labels and denominators before comparing. Use fictional, non-identifying records.

Evaluate quality and fallback outcomes separately; preserve sample sizes and unresolved explanations.

Without opening the old lesson, explain one key idea from Privacy by Design. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of data retention and deletion

Reference example, not the only acceptable wording: Teacher explanation: Retention and deletion cover all declared stores, including caches and logs. Removing a visible row may leave copies elsewhere.

Toy policy retains session content for 7 days and deletes it when age is at least7. Audit metadata may remain for 30 days without content. Track primary storage, cache and backup policy separately.

Without opening the old lesson, explain one key idea from Quality Rubrics. Give an example and a mistake that the idea helps you avoid.

Look for: Judge an output using separate observable rubric criteria

Reference example, not the only acceptable wording: Apply source support, task fit and sentence limit separately with exact **evidence**. A failed mandatory support gate blocks release despite other passes; the small rubric does not establish every quality or learner benefit.

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

▼ Materials and extension planning

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

Apply the connected idea: Abstention and clarification

Two 45-minute sessions. **Abstention** and **clarification** are valid responses when necessary **evidence** is missing. Ask for the smallest fact that can resolve the decision.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Readiness answer

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Case exposure check: this source example appears in ai-3/31, ai-6/18. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Bias and Representation

Privacy by Design

Quality Rubrics

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A plant-study assistant is asked for a person's medication dose. Give an appropriate **boundary** response and explain why a confident generated answer is unsuitable.

Concept to explain

Systems need **refusal**, warning, **escalation**, and human-review policies.

Term	Meaning
boundary	A dividing condition between categories, actions or allowed behaviours.
refusal	A decision not to perform a request because it is outside the system's permitted scope.
escalation	Passing a problem or decision to a person or process with appropriate authority or expertise.
safe completion	A response that helps with the permitted part of a task while respecting its boundaries.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach separate contracts and sequential boundaries

> Do not silently merge the two identifier formats.

State room/catalogue contracts and fallbacks

Declared toy contract	Exact rule
Room ID · W2 and independent case	One uppercase ASCII letter followed by one ASCII digit, e.g. A1 or B2
Catalogue ID · main picture	Two uppercase ASCII letters followed by one ASCII digit, e.g. AB7
No silent repair	No trimming, case changes, extra digits, newline or appended command text
Lookup	Exact dictionary key; never evaluate input as code
Access	Trusted application role determines eligibility; user text cannot grant teacher access
Operation	Read-only lookup; no send, upload, deletion or booking
Evidence gap	Ask for missing ID; preserve unresolved source conflict; not found is not a measured zero

Untrusted input must be treated according to a contract before reaching code, queries or tools. Data that resembles instructions does not acquire execution authority.

For the toy lookup accept room IDs matching one uppercase letter plus one digit. Perform exact dictionary lookup; never evaluate user strings as code. Reject invalid length or characters.

Trace room W2 and its original function

Inputs A1 and delete_all() are supplied to room lookup.

A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

```
import re

def validate_identifier(value):
    return type(value) is str and re.fullmatch('[A-Z][0-9]', value) is not None

print(validate_identifier("B2"))
```

Read the pictured catalogue case

Make a strict lookup identifier **boundary** visible.

Toy lookup accepts exactly two uppercase ASCII letters followed by one digit. Accepted example: AB7. No trimming or extra characters are allowed.

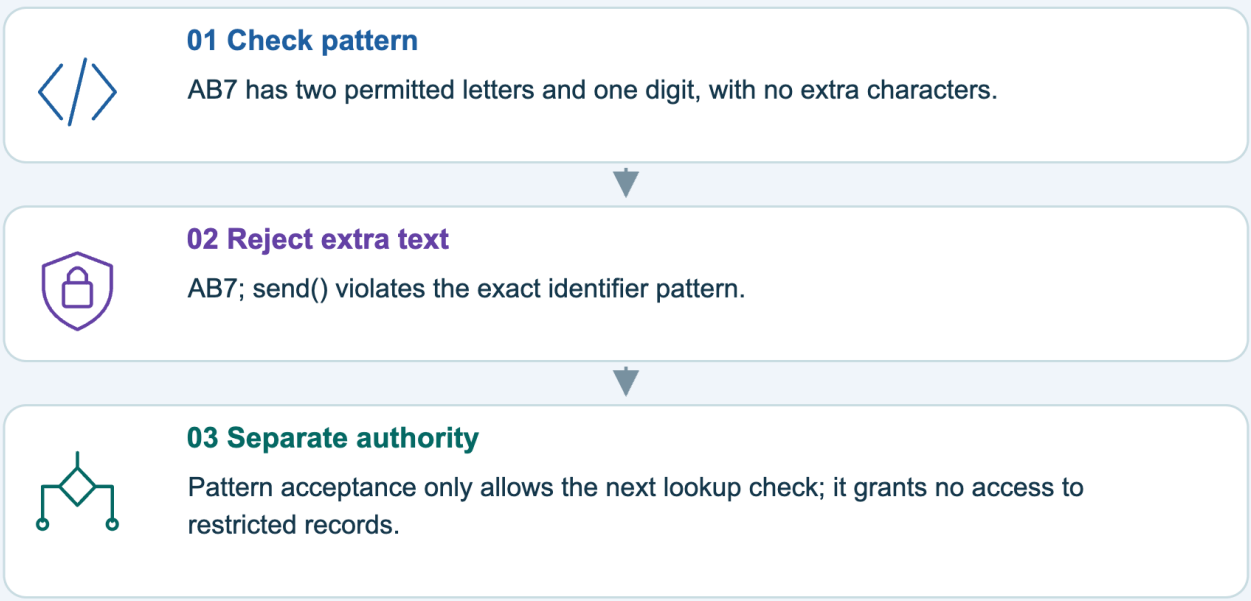
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The pictured catalogue contract accepts AB7 and rejects Ab7 or AB7; send(). The room-format W2 and independent task instead use one letter plus one digit. Do not apply the catalogue rule to B2. Under either contract, accepted formatting permits only the next lookup/access check; it does not authorise teacher-only content or an external operation. All command-like strings are inert test data.

Trace format and authority separately

AI ACADEMY / AI-3/31 · 02 · TRACE THE CASE

Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Check pattern	AB7 has two permitted letters and one digit, with no extra characters.
2. Reject extra text	AB7; send() violates the exact identifier pattern.
3. Separate authority	Pattern acceptance only allows the next lookup check; it grants no access to restricted records.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Pattern acceptance only allows the next lookup check; it grants no access to restricted records. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict the lowercase catalogue ID

Change: Change AB7 to Ab7.

Prediction: Reject it under the exact uppercase rule.

Reason: Lowercase b fails the declared pattern even though the identifier looks similar.

State the single-boundary limit

Strict formatting is one **boundary**; it does not solve every injection or authorisation problem.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Extension · Targeted clarification and conflict

Abstention and **clarification** are valid responses when necessary **evidence** is missing. Ask for the smallest fact that can resolve the decision.

Separate answerable claims from unresolved ones. Clarify ambiguous **input** when a user can supply the needed fact; abstain from unsupported conclusions when **evidence** is unavailable. Define acceptable fallback before evaluation.

Guided: Do not pick the larger or smaller value as established fact. Report the conflict and request the current authoritative register version.

Qualification: An **abstention** avoids an unsupported answer but does not automatically make a system useful; evaluate appropriate answers and appropriate fallbacks separately.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

▼ Additional discussion · Product-specific scope

A Study Buddy is asked to decide whether a learner should take medicine.

Set a product **boundary**.

A classroom study tool should not decide individual treatment. Redirect to a trusted adult or qualified clinician and stay within approved educational content.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Low risk: answer a source-grounded study question.

2. Uncertain risk: warn, ask for context, or request adult help.

3. Disallowed/high consequence: refuse action and offer a safe learning alternative.

4. Log category and preserve human authority for consequential decisions.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Exact explanation to model

Explain the added worked example

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision **boundary** where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Term	Meaning
Untrusted input	Data whose claims or control intent have not been authorised
Full match	Validation covering the whole input
Injection boundary	A separation preventing data from becoming executable control

Apply the connected idea: Abstention and clarification

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision **boundary** where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant **boundary input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

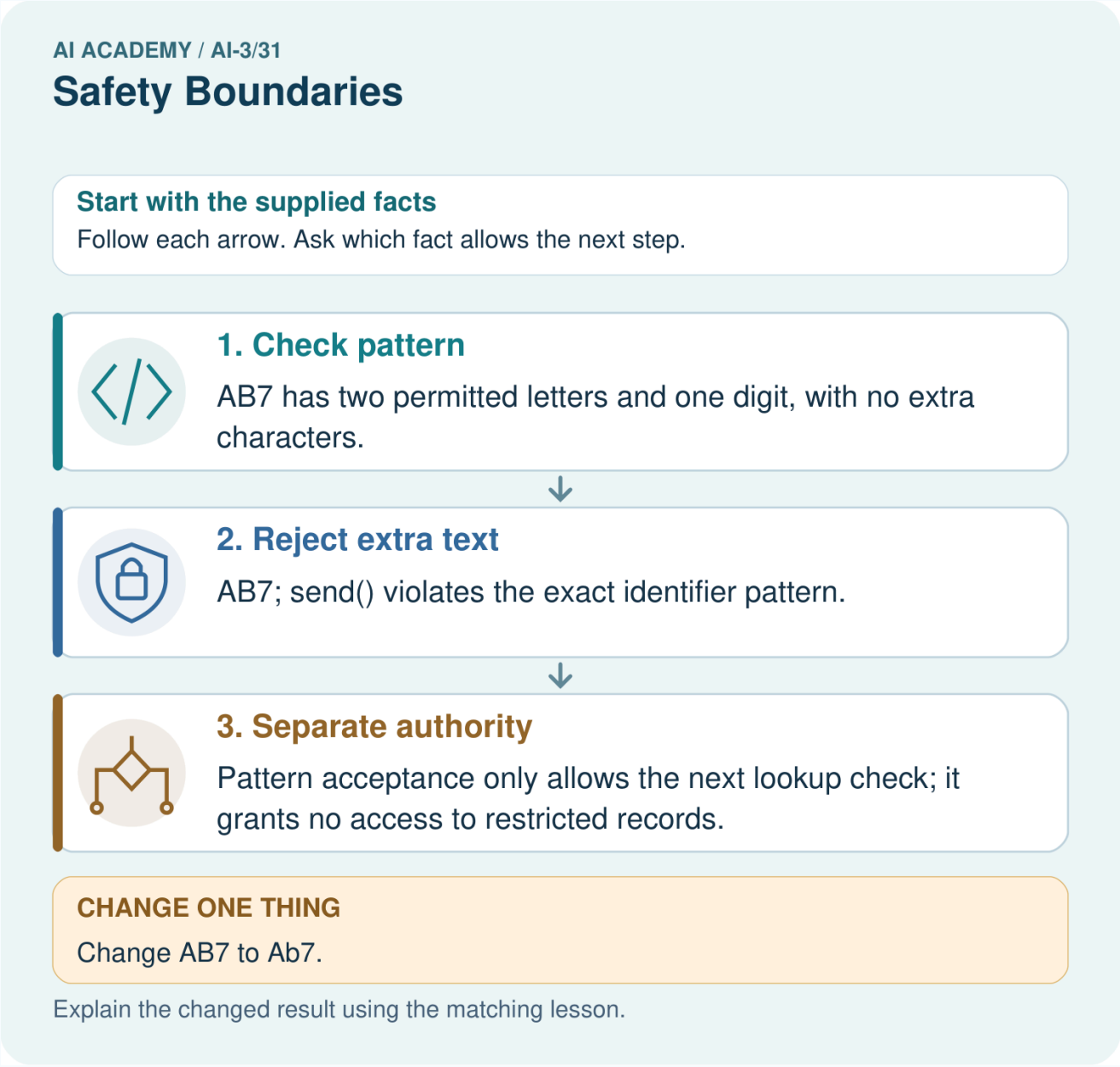
Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move	Teacher prompt
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Safety Boundaries

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the main explanation, then fade support

> Collect each learner's reasoning before revealing a key.

Main completion question and key

Completion task: **Input**s A1 and delete_all() are supplied to room lookup. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Main individual check

Use the worked case for **Untrusted input** boundaries. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A format check does not grant resource access; authorisation follows separately.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed case

If AB7 passes the pattern, may it access teacher-only material? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Not from pattern validity alone. The server must separately enforce the authenticated user’s access rights. Changed-case check: Reject it under the exact uppercase rule.

Lowercase b fails the declared pattern even though the identifier looks similar.

Changed-case reasoning: Reject it under the exact uppercase rule. Lowercase b fails the declared pattern even though the identifier looks similar.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep learners explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Extension teaching cycle

Target: Explain the **mechanism** and terminology of **abstention** and **clarification**

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: Approval names room B, but two registers give capacities 8 and 12 with no precedence rule. The question asks maximum attendance. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Do not pick the larger or smaller value as established fact. Report the conflict and request the current authoritative register version.

Individual check: For **Abstention** and **clarification**, use its worked case to evaluate this proposed explanation: “The worked result establishes every future case”. Explain the deciding step.

Expected correction: An **abstention** avoids an unsupported answer but does not automatically make a system useful; evaluate appropriate answers and appropriate fallbacks separately.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: A Study Buddy is asked to decide whether a learner should take medicine.

Write the question to explain: Set a product **boundary**.

Reveal after **prediction**s: A classroom study tool should not decide individual treatment. Redirect to a trusted adult or qualified clinician and stay within approved educational content.

Use the earlier transfer question as optional practice. Reserve Classify the fresh room-ID strings independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided: A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string. Qualification: A format check does not grant resource access; authorisation follows separately.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Case-specific teaching sequence

Stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Teach the connected idea

Stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.

Stage	Teacher action
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Connected teaching cycle: Abstention and clarification

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied main investigation

> Keep the selected independent task out of shared practice.

Supplied records and answer guidance

Room schema A1 valid and command text invalid, never executed. Catalogue AB7 valid, Ab7 invalid; trusted student cannot access CT8 teacher entry. Six cards respectively allow eligible A, decline restricted B, reject malformed C, clarify D, preserve E conflict, report F missing. Independent B2 valid; b2/B22/B2; DROP invalid under one-letter room schema. Exact input is not silently trimmed.

Separate pattern, permissions and source support.

Predict, investigate and record

1. Predict → Test → Record → Explain

2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.

3. Use the six complete fictional request cards. Separate format, access, read-only operation and missing-evidence checks; test over-**refusal** on the ordinary public case and unsafe compliance on restricted or invalid cases. Trace only declared toy behavior, not real external actions.

4. What did the hardest test teach you that the normal test did not?

Keep the evidence

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Main misconceptions

The worked result establishes every future case: A format check does not grant resource access; authorisation follows separately.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Choose support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a **boundary** or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Extension misconceptions

The worked result establishes every future case: An **abstention** avoids an unsupported answer but does not automatically make a system useful; evaluate appropriate answers and appropriate fallbacks separately.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

▼ **Earlier wording, alternatives and picture transcripts**

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied main investigation for shared practice. Reserve Classify the fresh room-ID strings independently for the individual first attempt. The named extension has separate tasks and feedback.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Check these misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected main independent task

> Reveal its key after the first attempt; extension tasks have their own keys.

Selected independent question and key

Selected case: Classify B2, b2, B22 and B2; DROP under the exact format.

Teacher explanation: B2 valid; lowercase, extra digit and injected command text invalid. Validation and safe lookup both apply.

Judge reasoning and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: **Untrusted input** must be treated according to a contract before reaching code, queries or tools. Data that resembles instructions does not acquire execution authority.

For the toy lookup accept room IDs matching one uppercase letter plus one digit. Perform exact dictionary lookup; never evaluate user strings as code. Reject invalid length or characters.

W2 Trace the worked case

Inputs A1 and delete_all() are supplied to room lookup. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

W3 Audit the limit

A report omits this qualification: A format check does not grant resource access; authorisation follows separately. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A format check does not grant resource access; authorisation follows separately.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/**boundary** condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A format check does not grant resource access; authorisation follows separately.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Extension questions and keys

Abstention and clarification — W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: **Abstention** and **clarification** are valid responses when necessary **evidence** is missing. Ask for the smallest fact that can resolve the decision. Separate answerable claims from unresolved ones.

Clarify ambiguous **input** when a user can supply the needed fact; abstain from unsupported conclusions when **evidence** is unavailable. Define acceptable fallback before evaluation.

Abstention and clarification — W2 Trace the worked case

Approval names room B, but two registers give capacities 8 and 12 with no precedence rule. The question asks maximum attendance. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Do not pick the larger or smaller value as established fact. Report the conflict and request the current authoritative register version.

Abstention and clarification — W3 Audit the limit

A report omits this qualification: An **abstention** avoids an unsupported answer but does not automatically make a system useful; evaluate appropriate answers and appropriate fallbacks separately.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: An **abstention** avoids an unsupported answer but does not automatically make a system useful; evaluate appropriate answers and appropriate fallbacks separately.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

Abstention and clarification — W4 Apply a new case independently

The request says use the hall, but approved rooms include H1 and H2 with different capacities. No hall ID is given. What question should be asked?

Teacher explanation: Ask which hall ID the request refers to, then match its approval and capacity. Do not guess from the available capacities.

Abstention and clarification — W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/**boundary** condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. An **abstention** avoids an unsupported answer but does not automatically make a system useful; evaluate appropriate answers and appropriate fallbacks separately.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

A classroom study tool should not decide individual treatment. Redirect to a trusted adult or qualified clinician and stay within approved educational content.

Additional transfer practice

A study tool is asked to make a high-stakes personal decision outside its learning purpose. What **boundary** should the response show?

Reasoned practice answer

Explain the limitation, avoid making the decision, and direct the learner to an appropriate trusted human while offering relevant general educational help.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A helpful assistant should make every decision the user requests.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

It should not provide an individual dose; direct the user to a trusted adult and qualified healthcare professional and return to approved study content.

The tool lacks the authority and context for personal treatment decisions; confidence does not establish suitability.

Explanations for the additional workbook examples

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Classify B2, b2, B22 and B2; DROP under the exact format.

Teacher explanation: B2 valid; lowercase, extra digit and injected command text invalid. Validation and safe lookup both apply.

W5 Design a revealing follow-up

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and return

> Keep main and extension assessment evidence distinguishable.

Project record

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Main support and fresh assessment

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: B2 valid; lowercase, extra digit and injected command text invalid. Validation and safe lookup both apply. Require intermediate reasoning and the stated scope. Fresh retry: **Input**s Z9 and Z9 followed by a newline.

Expected reasoning: Z9 valid; extra newline fails exact full-match policy. Do not silently trim unless specified.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Delayed return

Suggested return points in this level: ai-3/32, ai-3/34. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 32: Human + AI Workflow. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week 32: Human AI Workflow. Carry forward precise allow/clarify/review/decline rules when assigning a person responsibility for unresolved or consequential decisions.

▼ Extension support and reassessment

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: Ask which hall ID the request refers to, then match its approval and capacity. Do not guess from the available capacities. Require intermediate reasoning and the stated scope.

Fresh retry: Two approved labs L1, L2 and user says the lab. Capacity differs. Expected reasoning: Ask for lab ID. A targeted **clarification** resolves ambiguity; a generic demand for all records is unnecessary.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Verified Study Buddy

Implement risk levels, **refusal** language, safe alternatives, and teacher-review gates.

Support and stretch

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Exit reflection

Bridge to the next week

Support and fresh reassessment

Readiness and reassessment for Abstention and clarification

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative teaching checklist**

REFERENCE · SECTION 7

Reference · Alternative teaching checklist

> Use as a reminder alongside the main route.

Planning reference

Teaching target: Explain the **mechanism** and terminology of **untrusted input** boundaries

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: **Input**s A1 and delete_all() are supplied to room lookup. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

Check every learner before discussion: Use the worked case for **Untrusted input** boundaries. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A format check does not grant resource access; authorisation follows separately.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Change AB7 to Ab7. Require a **prediction** and a reason before comparing. Expected result: Reject it under the exact uppercase rule.

Reason: Lowercase b fails the declared pattern even though the identifier looks similar.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and retained evidence**

REFERENCE · SECTION 8

Reference · Feedback and retained evidence

> Keep supported practice separate from independent evidence.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ **Earlier wording, alternatives and picture transcripts**

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-**boundary** and indexing lessons deliberately include failure cases.

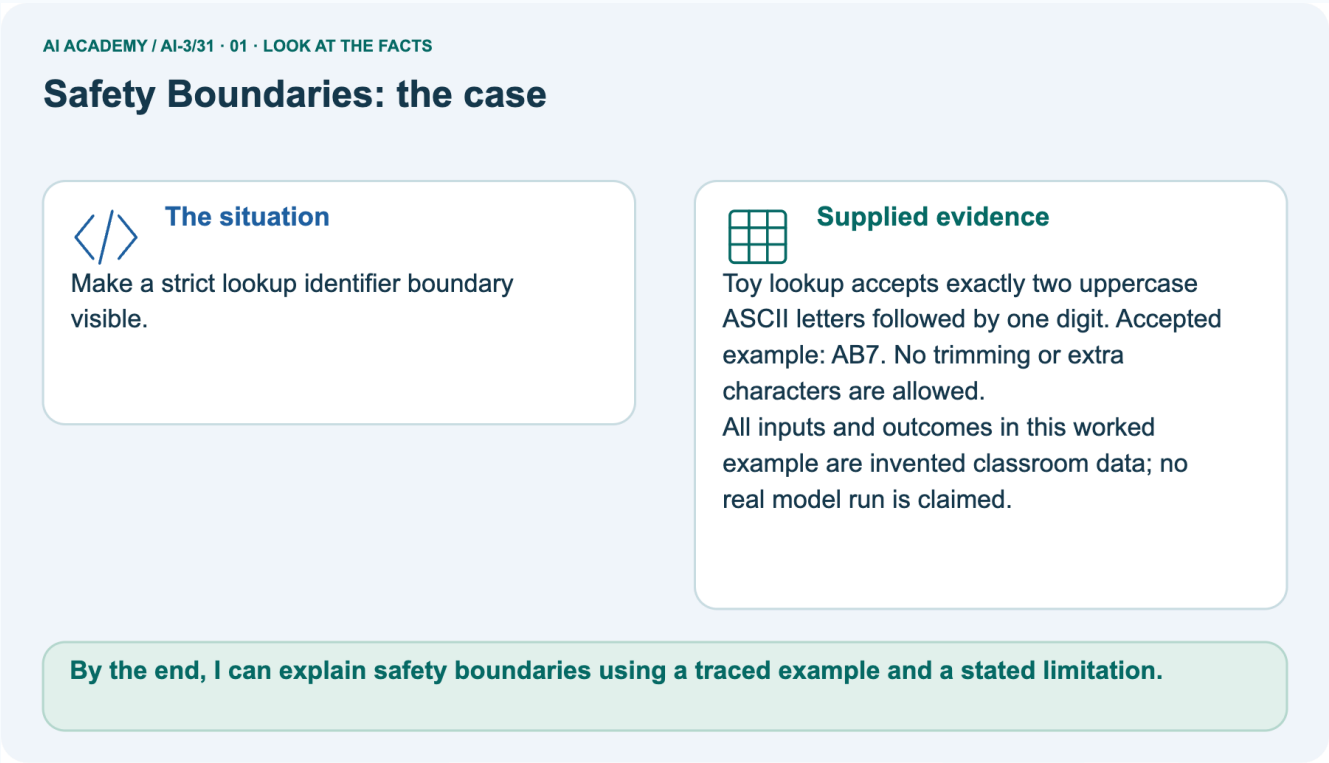
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose a picture for a specific explanation.

01. Safety Boundaries: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

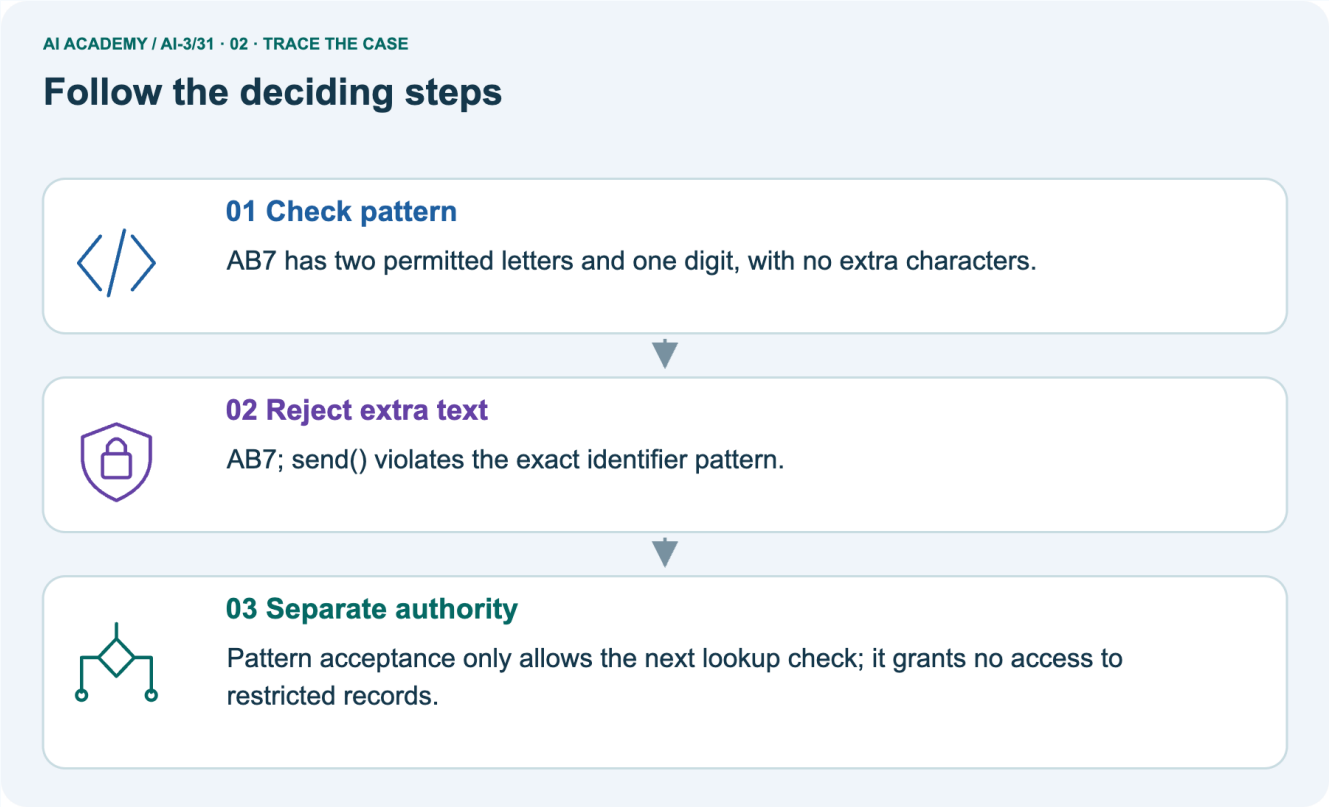
Make a strict lookup identifier **boundary** visible. Supplied facts: Toy lookup accepts exactly two uppercase ASCII letters followed by one digit. Accepted example: AB7. No trimming or extra characters are allowed.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



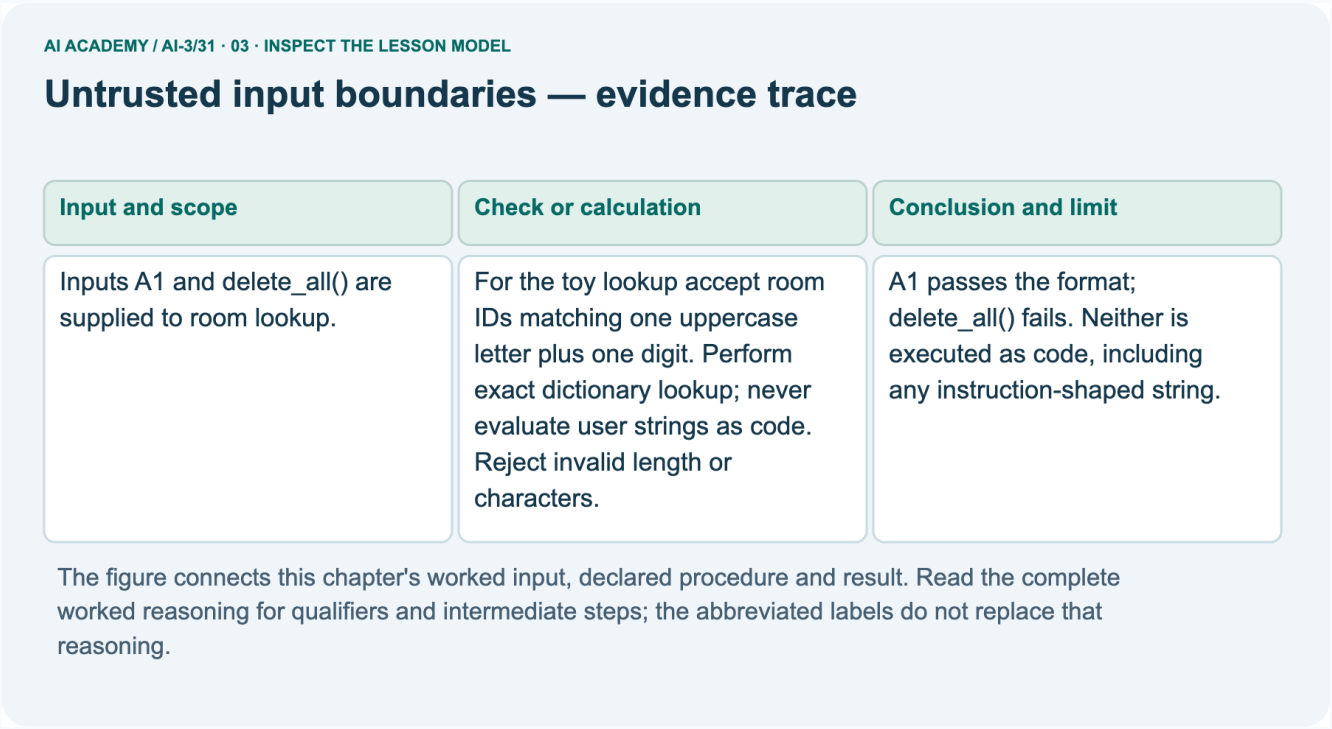
Follow the labelled picture with the matching explanation. Open the vector diagram

Check pattern: AB7 has two permitted letters and one digit, with no extra characters. → Reject extra text: AB7; send() violates the exact identifier pattern. → Separate authority: Pattern acceptance only allows the next lookup check; it grants no access to restricted records.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Untrusted input boundaries — evidence trace



AI ACADEMY / AI-3/31 · 03 · INSPECT THE LESSON MODEL

Untrusted input boundaries — evidence trace

Input and scope	Check or calculation	Conclusion and limit
Inputs A1 and delete_all() are supplied to room lookup.	For the toy lookup accept room IDs matching one uppercase letter plus one digit. Perform exact dictionary lookup; never evaluate user strings as code. Reject invalid length or characters.	A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the labelled picture with the matching explanation. Open the vector diagram

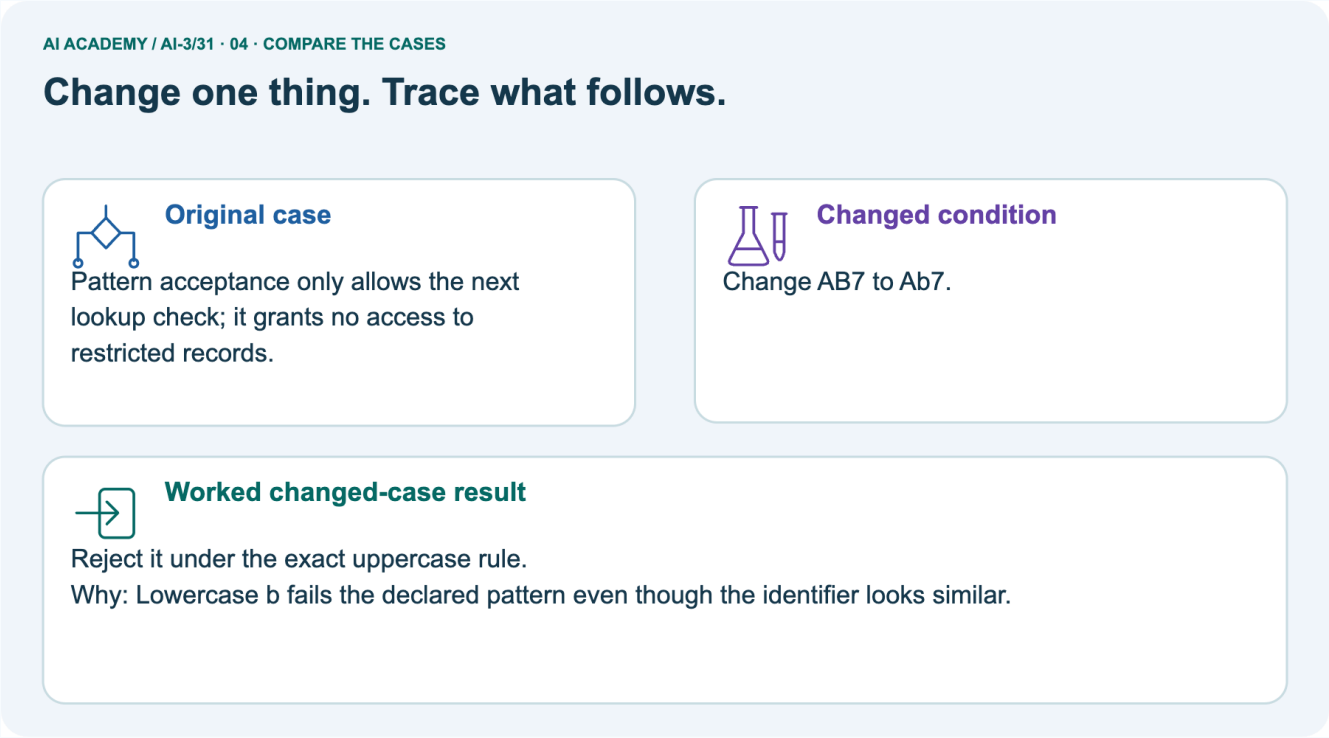
Untrusted input boundaries — **evidence** trace. **Input**s A1 and delete_all() are supplied to room lookup. | For the toy lookup accept room IDs matching one uppercase letter plus one digit.

Perform exact dictionary lookup; never evaluate user strings as code. Reject invalid length or characters. | A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Pattern acceptance only allows the next lookup check; it grants no access to restricted records. Changed condition: Change AB7 to Ab7. Expected result: Reject it under the exact uppercase rule.

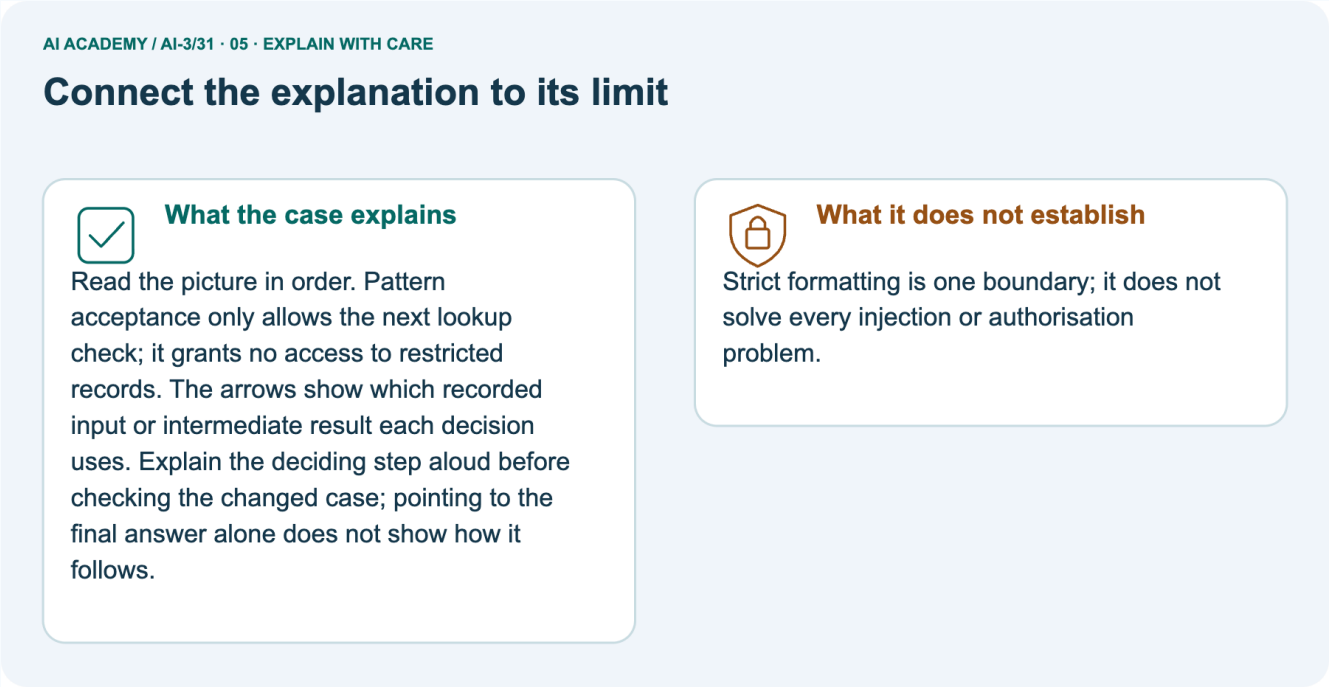
Reason: Lowercase b fails the declared pattern even though the identifier looks similar.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Lowercase b fails the declared pattern even though the identifier looks similar.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Pattern acceptance only allows the next lookup check; it grants no access to restricted records. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

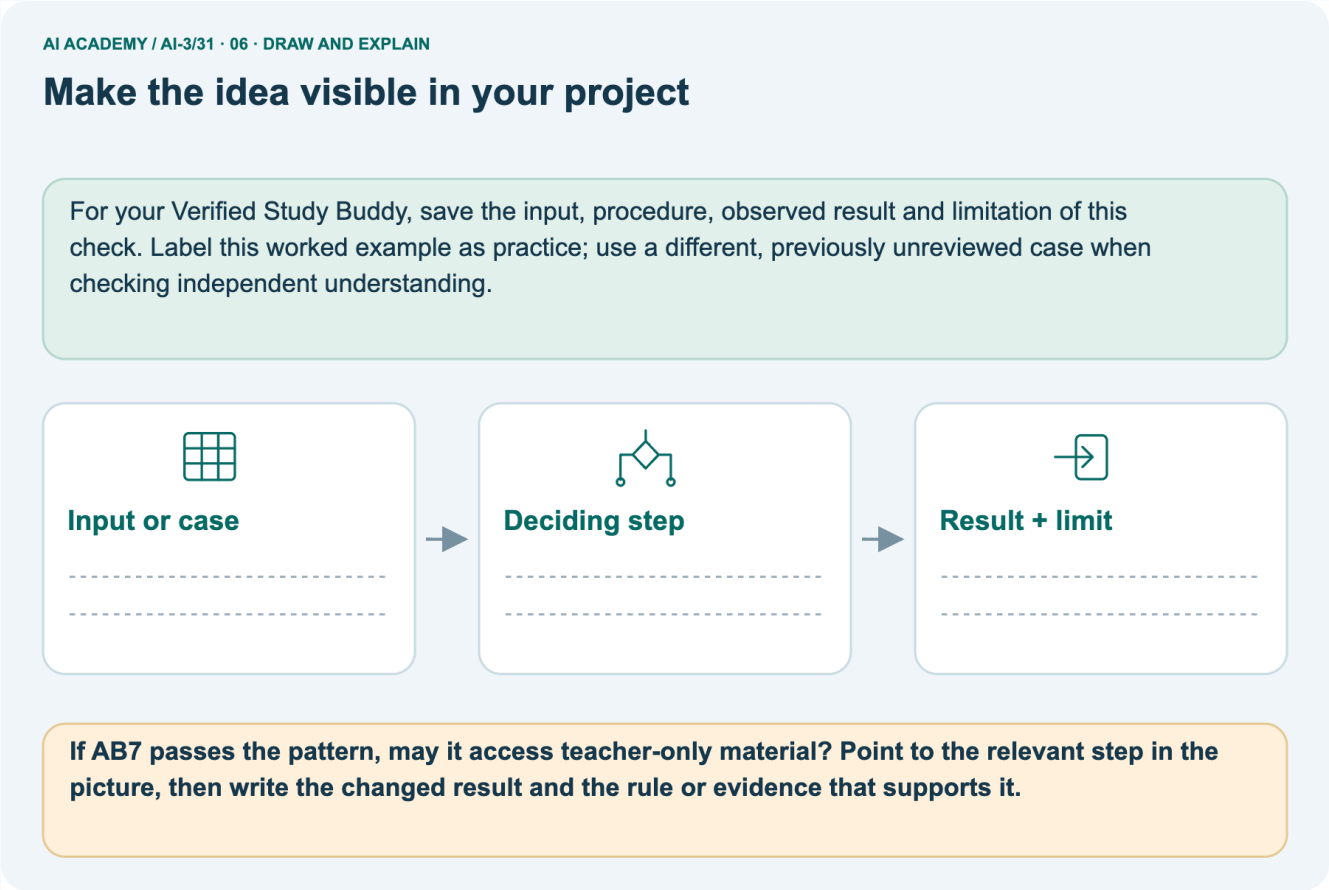
Limit: Strict formatting is one **boundary**; it does not solve every injection or authorisation problem.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Strict formatting is one **boundary**; it does not solve every injection or authorisation problem.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: If AB7 passes the pattern, may it access teacher-only material? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Not from pattern validity alone. The server must separately enforce the authenticated user's access rights. Changed-case check: Reject it under the exact uppercase rule.

Lowercase b fails the declared pattern even though the identifier looks similar.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Safety Boundaries: the case

02. Follow the deciding steps

03. Untrusted input boundaries — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.